

Labs





About the Course

This is the required lab component for Level 8 General Science. This course includes our custom laboratory book, which will build foundational and progressive skills and habits.

This topic is included in the following course(s): Science: Grade 8

About Science: Grade 8

In level 8, learners study science in its historical, political, and cultural context, and with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 8

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 8

In level 8, learners study science in its historical, political, and cultural context with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

Labs: Grade 8

The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	Labs: Grade 8 1 time/week 45 min	Science: Grade 8 Lab Book

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 8				



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 8

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Labs: Grade 8

- ☐ Preview science labs and purchase materials or have students gather them. Print or shortcut any links, as desired.

Special Topics & Field Trips

Science: Grade 8

Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

- ☐ Learn a few of your local species or varieties in connection with special topics.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

- ∞ [View Book List Details](#)

Labs: Grade 8



Science: Grade 8 Lab Book



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- ∞ [View Basic Supplies](#)
- ∞ [View Supply List Details](#)

Labs: Grade 8



Small Collection Containers



Scale/Balance



Quick Links

Labs: Grade 8

- ∞ [Grade 8 Lab Book](#)
- ∞ [Lab Notebook Examples](#)

Related Course Quick Links

- ∞ [Outdoor Work](#)
- ∞ [Seek app from iNaturalist](#)
- ∞ [SkyView Lite for iOS](#)
- ∞ [SkyView Lite for Android](#)
- ∞ [Alternate One Year Physical Science Lesson Plans](#)
- ∞ [Foundations \(See Section 13: Science\)](#)

Click THIS text
or scan the QR
code for links.



Labs: Grade 8

How To Teach Labs



Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

- On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

- Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

“I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work.”

- Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

- After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

- If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.



Lab Procedure

- Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

- The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

- If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

- Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.



Analysis & Conclusions

- The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

- Similar to the prelab narration, the concluding narration is a chance to think about and put into

words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

“It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn’t sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface.”

- Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.
- Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

Labs: Grade 8

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 45m Labs: Grade 8 - Lesson 1

Measuring Starburst Candy

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Measuring Starburst Candy, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and Procedure this week.

★ TEACHER TIP

Students measure manufactured candy to learn about metric, accuracy, and precision. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.

WEEK 2 45m Labs: Grade 8 - Lesson 2

Measuring Starburst Candy

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 2 of Measuring Starburst Candy, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Analysis and Conclusions today.

★ TEACHER TIP

Inviting students to share their lab notebooks is a great way to engage discussion and keep up with their learning! They should be able to discuss the science and explain what they did.

WEEK 3 45m Labs: Grade 8 - Lesson 3

Galileo's Falling Balls

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Galileo's Falling Balls, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. As time permits, they may begin steps 1-3 of the Procedure.

★ TEACHER TIP

Students work with velocity and acceleration while practicing with metric and use of a spreadsheet.

WEEK 4 45m Labs: Grade 8 - Lesson 4

Galileo's Falling Balls

 Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Galileo's Falling Balls, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure this week.

WEEK 5 45m Labs: Grade 8 - Lesson 5

Galileo's Falling Balls

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 3 of Galileo's Falling Balls, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

Labs: Grade 8

[Click THIS text or scan the QR code for links.](#)



Term 1

Suggested day 3: Students complete the Analysis and Conclusions this week.

WEEK 6 45m Labs: Grade 8 - Lesson 6

Magic of Inertia Activity

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete the Magic of Inertia activity, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

★ TEACHER TIP

This is a brief activity that will be completed this week. Be sure to have them share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 7 45m Labs: Grade 8 - Lesson 7

Forces

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Forces, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. If time permits, they might complete step 1 of the Procedure.

∞ Video Link: Step 1b How-to (2:32)

★ TEACHER TIP

Students adjust variables to the ramp from Lab 2 to evaluate forces related to gravity, friction, etc. Expect students to refer back to the process of using a spreadsheet from Lab 2. Reaching back to apply what was previously learned in a new situation is important. Encourage and model for them.

WEEK 8 45m Labs: Grade 8 - Lesson 8

Forces

 Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Forces, as directed in the Lab Book.

Suggested day 2: Students work through the Procedure today. If they need to take a break and complete the Procedure next week, they should do so between steps as directed in the lab.

WEEK 9 45m Labs: Grade 8 - Lesson 9

Forces

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 3 of Forces, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: If students have not yet done so, they should complete the Procedure. Analysis and Conclusions today. If they need an extra week, take the time needed to finish. There should be plenty of time for the last activity in Week 11.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

Labs: Grade 8

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 10 45m Labs: Grade 8 - Lesson 10

Equal and Opposite Activity

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of the Equal and Opposite activity, as directed in the Lab Book. Students may complete this activity in 1 week, but may also spend more time.

★ TEACHER TIP

Students play with balls to observe Newton's 3rd law.

WEEK 11 45m Labs: Grade 8 - Lesson 11

Equal and Opposite Activity

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete the Equal and Opposite activity or use this time to catch up on other labs or lessons in this course. If you have not already, be sure to narrate the lab to your teacher by showing them your lab notebook.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 12 45m Labs: Grade 8 - Lesson 12

→ EXAMS

Answer question(s) related to course.

Questions will come from:
Grade 8 Lab Book

Labs: Grade 8

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 1 45m Labs: Grade 8 - Lesson 13

Under Pressure Activity

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete the Under Pressure activity, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

★ TEACHER TIP

Students evaluate pressure and volume in 3 common phases of matter. Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 2 45m Labs: Grade 8 - Lesson 14

A Degree of Challenge

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of A Degree of Challenge, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. If time permits, they may complete the Procedure up to step 2.

★ TEACHER TIP

Students test the relationship between volume and pressure. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.

WEEK 3 45m Labs: Grade 8 - Lesson 15

A Degree of Challenge

Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of A Degree of Challenge, as directed in the Lab Book.

Suggested day 2: Students work on the Procedure this week. If step 2 was completed last week, then they might also complete Analysis and Conclusions. If they are just beginning the Procedure this week, then they may take another week to finish the lab.

WEEK 4 45m Labs: Grade 8 - Lesson 16

A Degree of Challenge

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 3 of A Degree of Challenge, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the lab this week. Some students who finished last week might jump ahead to the next lab.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 5 45m Labs: Grade 8 - Lesson 17

Elements and Reactions

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Elements and Reactions, as directed in the Lab Book.

★ TEACHER TIP

Students evaluate the reaction of iron and oxygen when steel wool rusts.



Term 2

Suggested day 1: Students complete the Introduction and begin the Procedure this week. There will be a wait of at least 1 week following step 7.

WEEK 6 45m Labs: Grade 8 - Lesson 18

Elements and Reactions

 Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Elements and Reactions, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure.

WEEK 7 45m Labs: Grade 8 - Lesson 19

Elements and Reactions

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 3 of Elements and Reactions, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Procedure and Analysis and Conclusions.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 8 45m Labs: Grade 8 - Lesson 20

Moving Particles, Electrical Energy, Magnetic Force

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Moving Particles, Electrical Energy, and Magnetic Force, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure this week.

★ TEACHER TIP

Students experiment with wire to observe a relationship between electricity and magnetism.

WEEK 9 45m Labs: Grade 8 - Lesson 21

Moving Particles, Electrical Energy, Magnetic Force

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 2 of Moving Particles, Electrical Energy, and Magnetic Force, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure and Analysis and Conclusions.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

Labs: Grade 8

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 10 45m Labs: Grade 8 - Lesson 22

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Faraday's Discovery, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure.

★ TEACHER TIP

Students produce electricity using a magnetic field.

WEEK 11 45m Labs: Grade 8 - Lesson 23

Faraday's Discovery

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 2 of Faraday's Discovery, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure and Analysis and Conclusions.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 12 45m Labs: Grade 8 - Lesson 24

→ EXAMS

Answer question(s) related to course.

Questions will come from:
Grade 8 Lab Book



Term 3

WEEK 1 45m Labs: Grade 8 - Lesson 25

The Photoelectric Effect

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of The Photoelectric Effect, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction.

★ TEACHER TIP

Students build a device to detect electric charge and observe how light interacts. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.

WEEK 2 45m Labs: Grade 8 - Lesson 26

The Photoelectric Effect

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 2 of The Photoelectric Effect, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure and Analysis and Conclusions today.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 3 45m Labs: Grade 8 - Lesson 27

Modeling Atoms

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Modeling Atoms, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction.

★ TEACHER TIP

Students create a model to convey what they know about the atom. Teachers should verify that students have what they need to begin their model next week.

WEEK 4 45m Labs: Grade 8 - Lesson 28

Modeling Atoms

Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of Modeling Atoms, as directed in the Lab Book.

Suggested day 2: Students work on their model.

WEEK 5 45m Labs: Grade 8 - Lesson 29

Modeling Atoms

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 3 of Modeling Atoms, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook and model.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.



Term 3

Suggested day 3: Students complete their model and the Analysis and Conclusions section.

WEEK 6 **45m Labs: Grade 8 - Lesson 30** *Measuring the Speed of Light Activity*

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of the Measuring the Speed of Light activity as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. They may begin the Procedure, as time permits.

★ TEACHER TIP

Students determine the speed of light empirically using a microwave.

WEEK 7 **45m Labs: Grade 8 - Lesson 31** *Measuring the Speed of Light Activity*

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 2 of the Measuring the Speed of Light activity as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure, as well as the Analysis and Conclusions section.

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 8 **45m Labs: Grade 8 - Lesson 32** *Interchanging Mass and Energy Practice*

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete Interchanging Mass and Energy Practice, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

If you need some help AFTER you have tried on your own, check out the video link:

∞ Video Link: Lab 4 Einstein (14:21)

★ TEACHER TIP

Students consider the meaning of $E=mc^2$ by using the equation. Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 9 **45m Labs: Grade 8 - Lesson 33** *What Hubble Saw*

 Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of What Hubble Saw, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction today.

★ TEACHER TIP

Students consider the meaning of 'red shift' using Hubble's calculations. Ideas concerning the expansion of the universe are discussed. Teachers might choose to allow this to pass by the way, to discuss with students, or to consider alternative theories they would like to share.

Labs: Grade 8

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 10 45m Labs: Grade 8 - Lesson 34

What Hubble Saw

Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY

Complete day 2 of What Hubble Saw, as directed in the Lab Book.

Suggested day 2: Students work on the Procedure today.

WEEK 11 45m Labs: Grade 8 - Lesson 35

What Hubble Saw

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 3 of What Hubble Saw, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Procedure and Analysis and Conclusions.

Stretching your imagination is much more important than getting the right answers, but if you need some help, you can check out the video link:
∞ Video Link: Lab 5 Einstein (7:53)

★ TEACHER TIP

Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 12 45m Labs: Grade 8 - Lesson 36

→ EXAMS

Answer question(s) related to course.

Questions will come from:
Grade 8 Lab Book